⑤ ECC vs RSA comparison for IOT Devices.

1. Given Info:
   RSA Key size = 1024 bits
   ECC Key size = 160 bits
   Device = IOT device

2. Computational cost comparison :-

Parameter	RSA (1024-bits)	Ecc (160-bit)
Key size	1024 bits	160 bits
Key generation	Generally Slower	generally faster
Encryption Verification	fast with Suitable public exponents	Requires elliptical curve operations
Decryption Signing	Computationally expensive	generally efficient
Memory usage	Higher	Lower.
Power consumption	generally higher overall	generally low overall.

3. Power Memory Consumption

RSA :- a) Uses Larger keys and Larger mathematical operations
   a) Requires more memory to store keys and perform calculations.
   a) Can Consume more energy especially during private key operations.

ECC :-

a) uses Smaller keys and compact Cryptographic data
b) Requires ~~more~~ Less memory to Store keys and perform calculations.
c) Can consume ~~more~~ Less energy, especially during private key operations. Suitable
~~d) Consume~~ for IoT devices.

ECC is better for IOT devices because it uses Smaller Key, less memory generally Consumes less power. more effecient for devices with Limited processing capacity.

Conclusion :- ECC is preferred over RSA for resources Constrained IoT devices.

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① RSA Encryption Analysis :

Given : $p = 11,\ q = 13,\ e = 7,\ M = 9$

a) Compute n and $\phi(n)$

$$n = p \times q = 11 \times 13 = 143$$
$$\phi(n) = (p-1)(q-1)$$
$$\phi(n) = 10 \times 12 = 120$$

b) find private key d

$$ed \equiv 1 \pmod{\phi(n)}$$
$$7d \equiv 1 \pmod{120}$$
Since $7 \times 103 = 721 = 6(120) + 1$
$$\boxed{d = 103}$$

c) public Key $(e, n) = (7, 143)$
d) private Key $(d, n) = (103, 143)$

c) Encrypt Message $M = 9$

$$C = M^e \bmod n$$

$$C = 9^7 \bmod 143$$

using modular exponentiation:

$$9^2 \bmod 143 = 81$$

$$9^4 \bmod 143 = 126$$

$$9^7 \bmod 143 = (9^4 \times 9^2 \times 9) \bmod 143$$

$$C = (126 \times 81 \times 9) \; 143$$

$$\boxed{C = 48}$$

d) decrypt & Verify

$$M = c^d \bmod n$$

$$M = 48^{103} \bmod 143$$

using modular exponentiation, the result is.

$$\boxed{M = 9}$$

How changing e affects Security

a) Small e : make encyption faster but poor design, implementation may be Vulnerable to attacks

a) Large e :- Generally increases encryption computation time.

a) Valid e :- Must satisfy $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$

a) Common choice :- $e = 65537$ is widely used in practical RSA because it offers effecient public Key operations.

Conclusion :- RSA encrypts 9 into 48 and decrypts it back to 9. Small primes are suitable only for educational purpose, not real world security.